

Insolvency Insight: Corporate Bankruptcy Prediction Using Machine Learning

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Abstract

Corporate bankruptcy prediction is an important problem in financial risk management. This project presents an end-to-end machine learning framework for predicting corporate bankruptcy using financial indicators obtained from historical company data. Multiple machine learning models were explored and compared, with XGBoost selected as the final predictive model due to its superior performance. The project incorporates data preprocessing, feature scaling, model evaluation, deployment using Streamlit, and version control using GitHub. The resulting web application enables users to perform bankruptcy risk assessment in a simple and interactive manner.

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1 Project Overview

InsolvencyInsight: Corporate Bankruptcy Prediction Using Machine Learning is an end-to-end machine learning project designed to predict corporate bankruptcy using financial indicators obtained from historical company records. Bankruptcy prediction plays a crucial role in financial risk management, investment decision-making, and corporate governance. Early identification of financially distressed firms allows stakeholders to take preventive measures and minimize potential losses.

The project utilizes machine learning techniques to analyze financial attributes and classify companies into bankrupt and non-bankrupt categories. In addition to model development, a web-based application was created using Streamlit, enabling users to interact with the predictive system through an intuitive interface.

2 Objectives

The primary objectives of this project are:

- To analyze financial indicators associated with corporate bankruptcy.
- To develop predictive machine learning models for bankruptcy classification.
- To compare multiple classification algorithms and identify the best-performing model.
- To deploy the final model through a web application.
- To provide an accessible decision-support tool for financial risk assessment.

3 Dataset Description

The dataset consists of financial indicators collected from Taiwanese companies. Each observation represents a company and contains numerous financial ratios and performance metrics. The target variable indicates whether the company eventually became bankrupt.

The dataset was examined for missing values, feature distributions, and class imbalance. Appropriate preprocessing techniques were applied before model training.

4 Data Preprocessing

The following preprocessing steps were performed:

1. Data cleaning and validation.
2. Identification and handling of missing values.
3. Feature scaling using StandardScaler.
4. Separation of predictor variables and target labels.
5. Train-test data splitting.

These preprocessing procedures ensure that the machine learning algorithms operate efficiently and produce reliable predictions.

5 Machine Learning Models

Several machine learning algorithms were evaluated to determine the most suitable model for bankruptcy prediction.

Table 1: Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	89.33%	31.48%	85.00%	45.95%	82.90%
Random Forest	93.87%	44.00%	55.00%	48.89%	94.31%
XGBoost	94.40%	48.39%	75.00%	58.82%	95.94%

Replace the values marked as XX

6 Model Evaluation

The predictive performance of the developed models was evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. The results indicate that the XGBoost classifier achieved the best overall performance among the considered machine learning algorithms.

Logistic Regression achieved high recall (85.00%), demonstrating its ability to identify a large proportion of bankrupt companies. However, its precision was relatively low, indicating a higher number of false positives.

Random Forest improved overall accuracy and ROC-AUC compared to Logistic Regression, showing better balance between classification performance and model robustness.

XGBoost outperformed all other models, achieving an accuracy of 94.40%, an F1-Score of 58.82%, and a ROC-AUC score of 95.94%. These results indicate that XGBoost provides the most reliable classification performance for bankruptcy prediction and was therefore selected as the final deployment model.

Table 2: Performance Metrics of the Final XGBoost Model

Metric	Value
Accuracy	94.40%
Precision	48.39%
Recall	75.00%
F1-Score	58.82%
ROC-AUC	95.94%

7 System Architecture

The workflow of the proposed system is illustrated below:

Dataset → Data Preprocessing → Feature Scaling → XGBoost Model → Prediction → Streamlit Application

The architecture combines data preparation, machine learning modeling, and web deployment into a unified framework.

8 Application Development

A web application was developed using Streamlit to make the prediction model accessible to end users. The application enables users to upload financial data and obtain bankruptcy predictions instantly.

Key features include:

- User-friendly interface.
- Real-time prediction generation.
- Automated preprocessing pipeline.
- Integration with the trained machine learning model.

The trained model and scaler were saved using the Joblib library and integrated into the Streamlit application for efficient deployment.

9 Deployment and Version Control

The project was managed using GitHub for version control and collaboration. All source code, notebooks, datasets, and deployment files were organized in a structured repository.

The application was deployed online using Streamlit Cloud, allowing users to access the system through a web browser without installing any software locally.

10 Results and Discussion

Experimental results demonstrate that machine learning techniques can effectively identify financially distressed firms. The XGBoost model achieved the best overall performance among the evaluated algorithms due to its ability to capture complex relationships between financial variables.

The developed system provides a practical framework for financial risk assessment and demonstrates the effectiveness of data-driven decision-making in bankruptcy prediction.

11 Future Scope

Several enhancements can be incorporated in future versions of the project:

- Integration of additional financial and macroeconomic variables.
- Application of deep learning techniques.
- Automated feature selection methods.
- Explainable AI techniques such as SHAP and LIME.
- Real-time financial data integration.
- Development of a dashboard for interactive analytics.

12 Conclusion

This project successfully demonstrates the application of machine learning to corporate bankruptcy prediction. Through data preprocessing, model comparison, XGBoost classification, and Streamlit deployment, an effective and user-friendly prediction system was developed. The project highlights practical applications of data science, machine learning, financial analytics, and cloud deployment, making it a valuable contribution to predictive financial risk assessment.